

Reranking by the Consumer’s Read: A Real Dissociation Whose Magnitude the Consumer Cannot Forecast

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Abstract

Vector search systems answer “what is close” with a fixed embedding metric. The downstream consumer of the results reads the embedding through its own operator, and the two need not agree. We rerank approximate-nearest-neighbor candidates by the consumer’s read metric $d_C(x, y)^2 = (x - y)^\top P_C (x - y)$ with $P_C = W^\top W$ taken from the consumer itself, never trained on retrieval outcomes, and measure what happens across sixteen consumer-corpus units on real moral-text and newsgroup substrates. Three findings. First, a conditional dissociation is real and replicates: task accuracy rises by up to +0.29 while agreement with the embedding-nearest neighbor collapses to 0.006–0.05, it appears only where the consumer is decodable and misaligned with the embedding geometry, and a matched consumer shows nothing. Second, two preregistered attempts to seal a quantitative law for the effect’s magnitude failed their own bars, and we report both failures as executed: per-unit conversion of measured headroom spans two orders of magnitude within a single sealed family, and neither published reliability weights nor cross-fitted classification headroom orders it. Third, a post-hoc study across all thirteen graded units identifies the magnitude in kind and shows why it resisted sealing: the gain equals the read metric’s own retrieval headroom at roughly unit slope (Pearson 0.91), but that quantity is not forecastable from the consumer’s calibration data at natural corpus sizes, where fold-level noise (0.01–0.06) is the size of the effects themselves. The value of consumer-relative retrieval is real, lawful in form, and invisible in advance to the consumer’s own calibration channel.

1 Introduction

Approximate-nearest-neighbor (ANN) retrieval fixes a metric when the index is built. FAISS ranks by L2 or inner product; production vector databases expose a short menu of global metrics and default to cosine. The application that consumes the retrieved items usually reads the embedding differently: a moderation model weighs a few directions heavily, a topic classifier weighs others, and neither read is the index metric. This paper asks what happens when the ranking step uses the consumer’s read instead. Candidates are generated by the index metric as usual; the top k are reranked by $d_C(x, y)^2 = (x - y)^\top P_C (x - y)$, where $P_C = W^\top W$ comes from the consumer’s own parameters. Nothing is learned against retrieval outcomes, and the index is not modified.

Three findings, in the order the evidence arrived. The dissociation is real and conditional: reranking by the read raises the consumer’s task accuracy while destroying agreement with the embedding-nearest neighbor, it requires the consumer to be both decodable from the embedding and misaligned with its geometry, and a consumer aligned with the geometry gains nothing. Our two attempts to preregister a quantitative law for the magnitude both failed their sealed bars, and the failures are part of the record reported here. The post-hoc account of why is the paper’s sharpest result: the magnitude equals the read metric’s own retrieval headroom at roughly unit slope, and that headroom cannot be estimated from calibration data at the corpus sizes natural consumers provide. A practitioner can obtain the gain. The consumer’s own calibration view cannot tell in advance how large it will be.

The components are classical, and we claim none of them. Mahalanobis and learned metrics, local metric learning, and trained rerankers are established; position-dependent metrics for exploratory data analysis go back to the learning-metrics program of Kaski and colleagues, and the view that embedding quality is relative to a user cost is explicit in NeRV. What this paper adds is the provenance of the metric, taken from the deployed consumer rather than learned for retrieval, and the measurement: the conditional dissociation with both arms sealed, the two honest failures, and the identification of the magnitude with an unforecastable potential.

2 Method

Substrate. Sentence embeddings from two frozen encoders: all-MiniLM-L6-v2 (384-d) over 20 Newsgroups for the construction sweep, and LaBSE (768-d, L2-normalized) for all graded units. Graded corpora are single-domain moral-text sets (Social-Chemistry-101 foundations and judgment, AITA privacy, hate-speech fairness, dark-pattern autonomy, Moral-Stories care, environmental claims), pinned at seal with SHA-256 manifests and row counts in the repository.

Consumers. Each unit is an (axis, corpus) pair. The consumer is a logistic probe on the embedding, $P_C = W^\top W$. Where a validated domain encoder exists for the axis, the probe is distilled from it: the encoder’s centroid-valence score labels the calibration split and the probe regresses that read onto the embedding space. Probes are trained on calibration data only, before grading data is opened.

Pipeline and metrics. FAISS IndexFlatL2 generates $k = 50$ candidates; reranking picks the top-1 by d_C ; the task scores the query’s label against the returned neighbor’s label. Reported per unit and seed: gain (reranked minus L2 accuracy), recall@1 (agreement with the L2-nearest), an isotropic control ($P_C = I$ must reproduce L2 within 0.005; it did, in every unit and seed of both families), and the oracle ceiling over the candidate set (at least 0.997 everywhere, so candidate generation never binds and every effect is the metric’s).

Discipline. Exploratory cells commit their predictions to version control before running. Claim-bearing runs require a sealed preregistration naming units, bars, seeds, and manipulation checks in advance; a one-day cooling-off separates family construction from sealing; graded verdicts are committed as executed, pass or fail. Family V1 sealed seven consumers with mechanical arm assignment (misaligned iff decodability $D \geq 0.75$ and headroom $H = D - A \geq 0.04$, where A is L2 1-NN accuracy; matched iff $|H| < 0.02$) and three disjoint graded seeds. Family V2 repaired V1’s two instrument defects found in its post-mortem: qualification moved to five-fold cross-fitting and every unit became single-corpus. All records, seals, and post-mortems are in the campaign repository.

3 The dissociation is real and conditional

The construction sweep established the shape. On synthetic embeddings with a forced off-axis label, reranking by the read moved task accuracy from 0.261 to 0.676 while recall@1 fell from 1.000 to 0.027. On real 20 Newsgroups embeddings with the consumer aligned to the task (topic), the effect vanished (-0.019): a well-trained encoder is already the topic consumer’s metric, and there is nothing to exploit. Between those poles the effect is monotone in measured misalignment: binarizing progressively lower-variance PCA directions of the same real embeddings moves the gain from $+0.08$ to $+0.41$. Natural attributes behave accordingly (document length $+0.185$; capitalization $+0.019$). A cross-lingual arm found that an English-trained read operator transfers to machine-translated Spanish and French databases with positive but attenuated gains ($+0.054$, $+0.033$), and refuted the stronger claim that the read survives translation better than L2 does.

The graded families confirm the qualitative law wherever it was tested. Across the thirteen graded units, eleven of the anisotropic consumer-seed cells show positive gain; the two matched-arm units sit at their predicted null (V1 privacy $+0.007$ mean; V2 autonomy $+0.0099$ against a 0.010 bar); recall@1 collapses in every unit; the isotropic control never moves. The largest sealed effects are large: $+0.263$ to $+0.292$ across three seeds on the care unit, and $+0.063$ to $+0.101$ on AITA privacy.

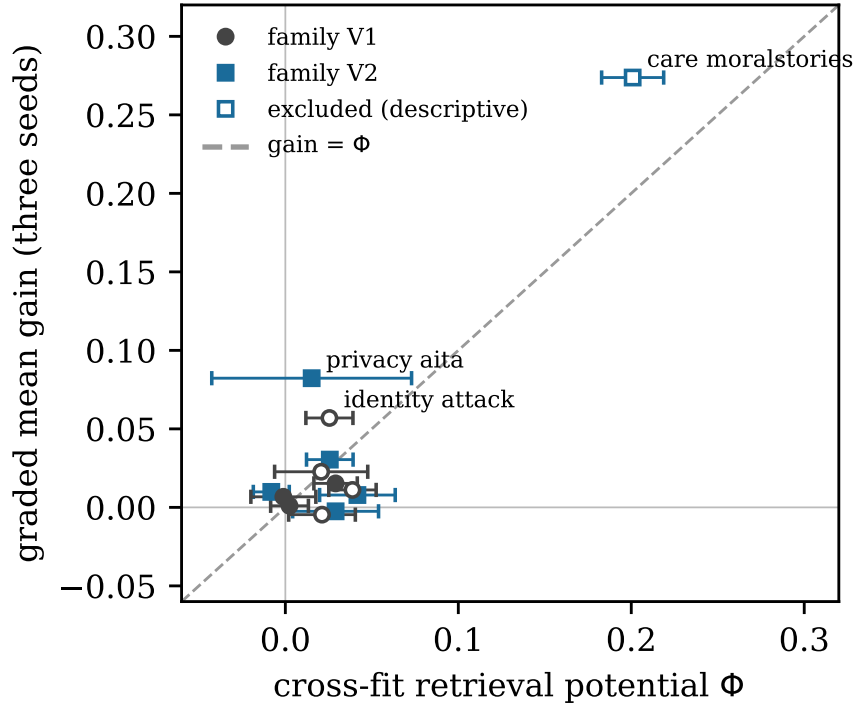


Figure 1: Graded mean gain against the cross-fit retrieval potential Φ , thirteen units, fold-SD error bars. One large- Φ unit sits on the unit line; the rest sit at the instrument’s noise floor. The correlation is Pearson 0.91 at slope 1.3, and the rank correlation is 0.22.

4 Two sealed failures, reported as executed

Family V1 barred each misaligned unit at $0.3H$ per seed and each matched unit at $+0.01$. Verdict: FAIL. Qualifying units converted about 9% of measured headroom against the sealed 30%; the matched bar broke by 0.003 on one seed. The post-mortem found two instrument defects. The single-split headroom estimate is noisy: one consumer was excluded at $H = +0.026$ when its cross-fit headroom was $+0.080 \pm 0.014$, and it then produced the run’s largest gains ($+0.053$ to $+0.062$). And pooled two-domain corpora manufacture headroom no query population carries: one axis was matched in one domain ($D = 0.944$, $A = 0.914$) and weak in the other ($D = 0.689$), its pooled $H = 0.048$ described neither, and its gain was correctly zero.

Family V2 repaired both defects and failed anyway, for a more instructive reason. With cross-fit qualification and single-domain units, per-unit conversion of measured headroom still spanned two orders of magnitude inside one sealed family: 190% (privacy), 74% (epistemic), 9% (fairness), ≈ 0 (environmental). The rank bar failed at Spearman 0.143. One unit exceeded its measured headroom outright, which shows the qualification quantity itself is the wrong potential: classification headroom $H = D - A$ bounds what the probe classifies, not what its metric retrieves. In both families the excluded row held the strongest confirmation: the V2 care unit, excluded for a weak probe ($D = 0.673$), converted 101% of a $+0.270$ headroom with three tightly clustered seeds. The qualification floor built to protect the misaligned arm from weak probes twice filtered out the largest effects of the phenomenon it guards.

5 The magnitude is the read’s headroom, and it is not forecastable

The post-hoc study asked what does order the magnitude. Three candidates fail: the published reliability weights of the source encoders (tested as V1’s secondary bar), classification headroom (Spearman 0.181

across the thirteen units), and single-split retrieval headroom refined by cross-fitting, which made the rank correlation worse (0.352 to 0.220). The quantity that succeeds in kind is the read metric’s own retrieval headroom: $\Phi = \text{xfit}(R_C - A)$, where R_C is 1-NN label accuracy under d_C on calibration data. Against Φ the graded gains lie near the unit line (Pearson 0.908, slope 1.30; Figure 1), and the identification is close to definitional once stated: with the oracle ceiling at 1 and k near-global, the graded gain is the grading split’s retrieval headroom under the frozen read.

Why then did two preregistrations fail? Because Φ cannot be estimated in advance at natural sizes. The fold-level standard deviation of Φ is 0.011–0.058 across units, and ten of thirteen measured gains are below 0.06: the effects sit at the noise floor of the only instrument the consumer has before grading. The one unit whose Φ stands eleven standard deviations from zero is forecast correctly (0.201 ± 0.018 predicted, 0.274 measured). Three units additionally show calibration-to-grading shift larger than their fold noise, so noise is not the only failure mode. A sealable forecast would need either engineered large- Φ units or calibration corpora an order of magnitude larger, and we record that requirement rather than attempt a third seal on the present pool.

The finding is worth stating in consumer-relative terms, because it is the same shape as the phenomenon it measures. What retrieval is worth depends on who reads the results; we add that how much it is worth is, at practical sizes, invisible to the reader’s own calibration channel. The two failed seals are the evidence for that sentence, and we regard them as load-bearing results rather than mishaps.

6 Related work

Metric learning, Mahalanobis and local variants, and trained rerankers are long established; our reranker is not learned and our claim is not a better reranker. Position-dependent Riemannian metrics from auxiliary information were developed for exploratory analysis in the learning-metrics program (Kaski et al., 2001; Peltonen et al., 2004), and NeRV made embedding quality explicitly relative to a user’s precision-recall cost (Venna et al., 2010); our P_C is read off a deployed consumer rather than built from labels at embed time, and where the consumer is itself a label probe the constructions nearly coincide. Task-aware and decision-focused pipelines optimize models for downstream decisions (Elmachoub and Grigas, 2022); sensor-selection and value-of-information literatures optimize weighted-trace objectives (Joshi and Boyd, 2009). We do not offer a better metric, reranker, or selection rule. We offer the measurement: the conditional dissociation with both arms sealed, two preregistered failures reported as executed, and the identification of the magnitude with a potential the consumer cannot estimate in advance.

7 Scope and limitations

One embedding family per phase, linear probes, top-1 label prediction, and $k = 50$ candidate generation whose ceiling never binds. The corpora are English moral text and newsgroups; the cross-lingual arm covers machine-translated Spanish and French databases only, and its stronger invariance claim is refuted above. The distillation rule ties three units to validated domain encoders; the rest use direct labels. All limits push in the same direction: they narrow the settings in which we have measured the law, not the honesty of the failures. In our controlled tests nothing suggested the dissociation appears without measured misalignment, and we claim nothing about that regime.

Reproducibility

Every number above is read from committed, hashed records in the campaign repository (sealed preregistrations, graded verdict JSONs, post-mortems, and the determinant study), and the figure regenerates from them by script. The two sealed families’ verdicts are immutable; the repository’s evidence discipline file describes the seal and amendment rules. An anonymized copy of the records accompanies the submission; the repository link is provided on acceptance.

References

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